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Interaction and inactivation of *Listeria* and *Lactobacillus* cells in single and mixed species biofilms exposed to different disinfectants

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Abstract

Listeria spp. are ubiquitously found in both the natural and the food processing environment, of which *Listeria monocytogenes* is of an important health risk. Here, we report on the formation of single and mixed species biofilms of *L. monocytogenes*/*Listeria innocua* and *Lactobacillus plantarum* strains in 24-well polystyrene microtiter plates and on the inactivation of 24-hr and 72-hr biofilms using quaternary ammonium compound-, tertiary alkyl amine-, and chlorine-based disinfectants. Fluorescent *in situ* hybridization (FISH) and LIVE/DEAD BacLight staining were applied for 72-hr *L. innocua*–*L. plantarum* mixed biofilms in the LabTek system for the species identification and the reaction of biofilm cells to disinfectants, respectively. *L. monocytogenes*/*L. innocua* were more resistant to disinfectants in 72-hr than in 24-hr biofilms, whereas *L. plantarum* strains did not show any significant differences between 72-hr and 24-hr biofilms. Furthermore, *L. innocua* when grown with *L. plantarum* was more resistant to all disinfection treatments, indicating a protective effect from lactobacilli in the mixed species biofilm. The biofilm formation and reaction to disinfectants, microscopically verified using fluorescence *in situ* hybridization and LIVE/DEAD staining, showed that *L. innocua* and *L. plantarum* form a dense mixed biofilm and also suggested the shielding effect of *L. plantarum* on *L. innocua* in the mixed species biofilm.

1 | INTRODUCTION

Today, the genus *Listeria* includes 17 species, of which *Listeria monocytogenes* poses a significant health risk (Orsi & Wiedmann, 2016). *L. monocytogenes* has been implicated in a number of food-borne outbreaks, causing infections in healthy subjects (Todd & Notermans, 2011); however it mainly affects individuals from the high-risk group, the so-called young, old, pregnant, immunocompromised (YOPI) (Jeyaletchumi et al., 2010). A *Listeria* species that is genetically, ecologically, and phenotypically close to *L. monocytogenes* is *Listeria innocua*, which is also being widely distributed in the environment and in food (Perrin, Bremer, & Delamare, 2003). Although *L. innocua* is considered to be a nonpathogenic bacterium, this species has already been associated with a fatal bacteraemia and an acute meningitis in individuals from YOPI group (Favaro et al., 2014).

Listeria cells spread by a wide range of ready-to-eat (RTE) foods (Cordano & Jacquet, 2009; Kramarenko et al., 2013; Slama, Bekir, Miladi, Noumi, & Bakhrouf, 2012) usually due to post-pasteurization contamination from the food processing environment. The source of listerial contamination of food might be the formation of biofilms as a result of the insufficient hygienization of the plant environment (Berrang, Meinersmann, Frank, Smith, & Genzlinger, 2005; Chaitiemwong, Hazeleger, & Beumer, 2010). Biofilms are complex communities of microorganisms in which cells adhere to surfaces and are embedded in an exopolysaccharide (EPS) matrix (Borucki, Peppin, White, Loge, & Call, 2003), serving as a protection against multiple environmental factors (Norwood & Gilmour, 2001). Consequently, the cleaning and disinfection treatments effective toward planktonic cells may turn out to be ineffective towards biofilms (Saá Ibusquiza, Herrera, Vázquez-Sánchez, & Cabo, 2012).

Therefore, in order to design an effective strategy for biofilm control, it is crucial to advance our understanding of the interaction between various antimicrobial agents and biofilm cells. It is also important to judge whether the antimicrobial behavior on biofilms depends on interactions between different species, since in a variety of environments including the plant environment, biofilms consist of multiple species. For instance, mixed species biofilms for *L. monocytogenes* and *Lactobacillus plantarum* have been studied by Van der Veen and Abee (2011), showing their increased resistance to benzalkonium chloride and peracetic acid. Indeed, both of them are encountered in similar environmental niches, and as such, they may be introduced into the facility with the same raw materials used by the food industry, posing a higher risk of food contamination. However, there are still few information on the inactivation and interaction between both species.

Therefore, the aim of this study was to evaluate the performance of *L. monocytogenes*/*L. innocua* and *L. plantarum* in single and mixed species biofilms and further to elucidate effects of three disinfectants: quaternary ammonium compounds (QACs)-based, tertiary alkyl amine-based, and chlorine-based, on these biofilms at different stages of their maturity. Biofilm formation characteristics, including antimicrobial behavior of disinfectants, were also evaluated by using the LabTek chamber slide system for biofilm development and epifluorescence microscopy (EFM) for biofilm visualization.

2 | MATERIALS AND METHODS

2.1 | Bacterial strains

Three *Listeria* strains: *L. monocytogenes* ATCC 19112 (reference strain), *L. monocytogenes* FI 1, and *L. innocua* FI 2 (food isolates, sweet paper); and three *Lactobacillus* strains: *L. plantarum* PC 1.1, *L. plantarum* PC 1.2, and *L. plantarum* PC 3.2 (food isolates, pickled cucumbers), were used in this study, because these six strains were found to be strong biofilm producers in our previous experiment with crystal violet (CV) staining and *n*-hexadecane assay, comprising 8 strains of *Listeria* and 10 strains of *Lactobacillus* (data not shown). All cell cultures were short-term stored at 4°C either on tryptic soy agar (TSA; Merck, Darmstadt, Germany) or De Man, Rogosa and Sharpe agar (MRS-agar; Merck).

2.2 | Single and mixed species cell cultures

Individual *Listeria* spp. and *L. plantarum* strains were grown in tryptic soy broth (TSB; Merck) at 37°C and in MRS (Merck) broth at 30°C, respectively, for 24 hr. Overnight single species cell cultures were adjusted to a cell density of 1×10^6 cells/ml in either TSB or MRS. In turn, equal volumes of the adjusted single species cell cultures were combined to prepare *Listeria*–*Lactobacillus* mixed species cell cultures. In a preliminary study, several broths: TSB, TSB with 0.005% manganese sulfate (Mn), brain heart infusion (BHI; Merck), and BHI with 0.005% Mn were tested to determine the optimum one that supports the growth of two species in a co-culture. Consequently, *L. innocua* FI 2 and *L. plantarum* PC 1.2, which showed the greatest contribution in a co-culture, were selected for experiments under mixed-species conditions in BHI with 0.005% Mn.

2.3 | Single and mixed species biofilm growth

The prepared single or mixed species cell cultures were added to 24-well flat-bottom polystyrene plates (Costar, Corning, NY) at 1 mL per well. The plates were incubated statically at 37°C and 30°C for *Listeria* spp. and *L. plantarum*, respectively, for 72 hr, while growing in single species biofilms and at 30°C for *Listeria*–*Lactobacillus* while forming mixed species biofilms. Sampling was performed at 24-hr time intervals by removing the supernatant fluid with planktonic cells. The remaining biomass of each well was harvested by scraping the plate surface with a sterile polyester-tipped swab (Deltalab Eurotubo, Barcelona, Spain). Swabs were placed in glass tubes containing 9 mL of saline (0.85% NaCl) and sonicated at 38 kHz (Elmasonic SB-120DN; Abchem, Warsaw, Poland) for 1 min followed by vigorous vortexing for another 1 min. Then, 100 μL of the biofilm-released cells or their appropriate dilutions in saline were spread plated in duplicate on TSA and MRS-agar plates for samples of single species biofilms and on Oxford-agar (BIOCORP, Warsaw, Poland) and MRS-agar plates for samples of *Listeria*–*Lactobacillus* mixed species biofilms. The plates were incubated at 37°C in the case of *Listeria* spp. and at 30°C under anaerobic conditions (AnaerocultC; Merck) in the case of *L. plantarum* for 48 hr. Following incubation, typical colonies of *Listeria* spp. and *L. plantarum* were counted and the results were calculated as log CFU/ml. The mean values were obtained from three independent trials with three repetitions for each trial (n = 9).

2.4 | Single and mixed species biofilm inactivation assay

Single and mixed species biofilms of *Listeria* spp. and *L. plantarum* were first developed as described earlier. The 24- and 72-hr grown biofilms were further tested for inactivation experiments with: quaternary ammonium compound-based sanitizer—Pursept-AF (12.5 g didecyl-dimethylammonium chloride, 1.5 g *N*-(3-aminopropyl)-*N*-dodecylopropane-1,3-diamine; Merz, Frankfurt, Germany) at 2.0% (as recommended for use by the manufacturer), tertiary alkyl amine-based sanitizer—Barren (<5% anionic surfactants, <5% EDTA and its salts; Amtra, Sosnowiec, Poland) at 2.0%, and a chlorine-based sanitizer—Medicarine (99% sodium dichloroisocyanurate; Ecolab, St. Paul) at 0.18% (1.000 ppm; as recommended for use by the manufacturer). All disinfectant solutions were prepared in distilled water according to the manufacturer's instructions. Sterile PBS (Sigma-Aldrich, Poland) was used as a control. First, medium was removed and the biofilms were gently rinsed with PBS. One milliliter of disinfectants or PBS (control) was added into the wells. After 5-min treatments, disinfectant solutions or PBS were removed and the wells were filled with 1 mL of a D/E neutralizing broth (BD, Le Point de Claix, France). After 10 min, the neutralizing broth was removed and wells were washed with PBS (Sigma-Aldrich) before enumeration with plate counting. All results were expressed as log reductions (–). The mean values were obtained from three independent trials with three repetitions for each trial ($n = 9$).

2.5 | FISH and cell viability in mixed species biofilms

Overnight cell cultures of *L. innocua* FI 2 and *L. plantarum* PC 1.2 (1:1, v/v) were equally adjusted to a cell density of 1×10^2 cells/ml and added to sterile four-well LabTek chamber slides (Thermo Fisher Scientific, Waltham, MA) in a final volume of 800 μL. The slides were incubated at 30°C for 72 hr. The medium was changed at 24-hr time intervals. After 72 hr, the biofilms were inactivated with the disinfectants described above and PBS was used as a control. After 5 min, the disinfectants or PBS were removed and the wells were filled with the D/E broth, incubated for 10 min and then washed with PBS. Next, the biofilms were stained according to the LIVE/DEAD protocol to determine their viability. One milliliter of the staining mixture (1.5 μL of component A—Syto9 and 1.5 μL of component B—PI, in 1 mL of sterile saline) was transferred into each chamber. The mixture was incubated at 37°C for 30 min in the dark. Afterward, it was removed, whereas the biofilms were washed with PBS and mounted on glass slides underneath cover slips with immersion oil (Molecular Probes). The assessment of biofilms viability was performed with an Olympus BX51 epifluorescence microscope (Olympus, Hamburg, Germany). The microscope was equipped with a 100 W mercury lamp, XC digital camera, and a set of filters: U-MNB2: 470–490 nm and U-MNG2: 530–550 nm for detection of green (Syto9) and red (PI) fluorescence, respectively. Thirty pictures were taken for each biofilm and biofilm adhesion was evaluated in a nine-stage adherence scale (Le Thi, Prigent-Combaret, Dorel, & Lejeune, 2001). For biodiversity assessment, fluorescence *in situ* hybridization (FISH) was performed on untreated biofilms. The first step in FISH was biofilm fixation in a PBS-ethanol mixture (1:1, v/v), which was then incubated at −20°C for 30 min. Afterward, the PBS-ethanol mixture was removed and the bacterial membrane were permeabilized using lysozyme (Merck). Lysozyme concentrations of 10, 15, and 20 mg/mL were checked in the preliminary study. Following optimization step, we selected the lysozyme concentration of 15 mg/mL. The permeabilization was performed at 37°C for 30 min. Then, the lysozyme was removed and the biofilms were prepared for the hybridization step. The oligonucleotide probe specific to *Listeria* species: 5' CCC CAA CTT ACA GGC 3′ (Brechm-Stecher, Hyldig-Nielsen, & Johnson, 2005) was marked with a Pacific Blue (PB) and the probe specific to *L. plantarum*: 5' CCG TCA ATA CCT GAA CAG 3′ (Ercolini, Hill, & Dod, 2003), with cyanine 3 (CY3). The probes were prepared in a hybridization buffer, separately. The hybridization was performed in a Falcon tube at 40°C for 4 hr in an incubator (GFL 3031; Burgwedel, Germany). The probe specific to *Listeria* was subjected to 2-hr hybridization. Then, the hybridization mixture was removed and the biofilms were washed with PBS. The next step of hybridization was performed with the probe specific to *L. plantarum*. Next, the hybridization mixture was removed and the biofilms were washed with a pre-warmed washing buffer at 48°C for 15 min. Then, the washing buffer was removed and the biofilms were washed with PBS. Finally, the LabTek chamber slides were covered with a coverslip with Citifluor AF1 (Citifluor Ltd., London, England) and VectaShield (Linaris, Wertheim, Germany) (4:1, v/v) and stored at −20°C until microscopic analysis. The analysis was performed with an Olympus BX51 epifluorescence microscope (Olympus) with a set of filters: U-MNU2: 360–370 nm and U-M41007: 530–560 nm for detection of blue and orange fluorescence, respectively.

2.6 | Statistical analysis

Statistical analysis was conducted using Statistica software ver. 13.1 (StatSoft Inc., Tulsa, OK). Inactivated samples were tested by one-way ANOVA test. The differences were considered significant at $p < .05$ level of probability.

3 | RESULTS

3.1 | Performance of individual strains in single and mixed species biofilms

The average plate count cell densities of *Listeria* spp. and *L. plantarum* strains in single and mixed species biofilms during static growth in polystyrene plates were presented in Figure 1. In the single species biofilms, the highest cell number was determined after 24 hr and reached 8.6, 8.5, and 8.4 log CFU/ml for *L. monocytogenes* ATCC 19112, *L. monocytogenes* FI 1, and *L. innocua* FI 2, respectively, as well as 8.4, 8.1, and 8.0 log CFU/ml for *L. plantarum* PC 1.1, *L. plantarum* PC 1.2, and *L. plantarum* PC 3.2, respectively. Cell counts were observed to decrease in the subsequent days, and after 72 hr their reduction exceeded 1 log unit for all *Listeria* spp. (7.3, 7.4, and 7.3 log CFU/ml) ($p < .05$) and for *L. plantarum* PC 1.1 and *L. plantarum* PC 1.2 (7.2 and 6.8 log CFU/ml). *L. plantarum* PC 3.2 reached its lowest reduction, resulting in the cell count at 7.3 log CFU/ml.

In the mixed species biofilms of *L. innocua* FI 2 and *L. plantarum* PC 1.2, their highest cell counts were determined after 24 hr, that is, 7.4 and 7.3 log CFU/ml, respectively. In the subsequent days, cell counts slightly decreased and after 72 hr reached 7.1 and 7.0 log CFU/ml, respectively, however not significant ($p > .05$).

3.2 | Inactivation of individual strains in single species biofilms

Results of cell inactivation in the single species conditions allowed observing a higher resistance of *Listeria* spp. to the tested disinfectants in the 72-hr biofilms (Figure 2). Cell reductions in the 24-hr and 72-hr biofilms treated with the QAC-, tertiary alkyl amine-, and chlorine-based disinfectants differed by 4.6, 2.4, and 2.1 log units, respectively, in the case of *L. monocytogenes* ATCC 19112 and by 5.1, 3.7, and 2.1 log units, respectively, in the case of *L. innocua* FI 2 ($p < .05$). *L. monocytogenes* FI 1 was also more resistant in 72-hr biofilms, but only in these treated with the tertiary alkyl amine- and chlorine-based disinfectants. Cell reductions in its 24-hr and 72-hr biofilms differed by 3.1 and 3.2 log units, respectively. In turn, its exposure to the QAC-based disinfectant resulted in a lower cell

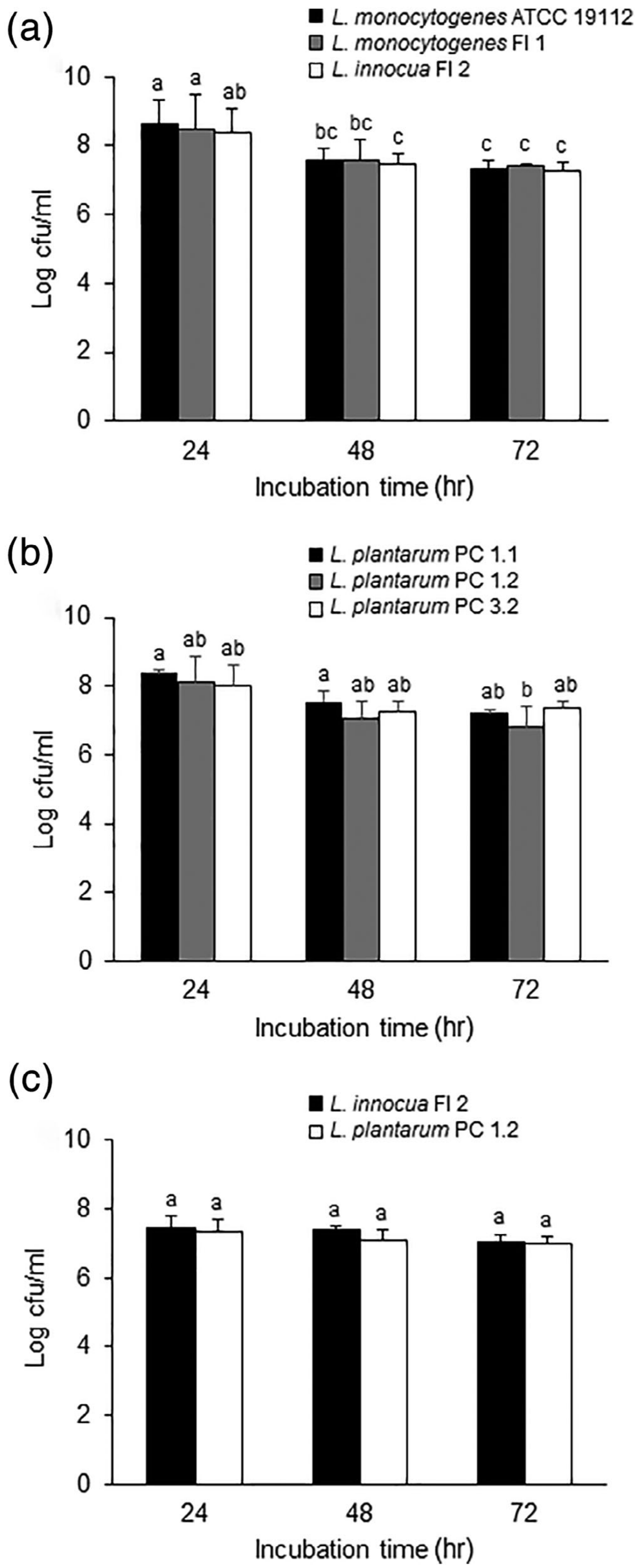

FIGURE 1 The cell counts of *Listeria* spp. and *L. plantarum* strains in single (a/b) and mixed (c) species biofilms formed in 24-well polystyrene plates at 24-hr time intervals. The bars represent the mean values ± *SDs* (n = 9). Different letters indicate significant differences between bars (p < .05)

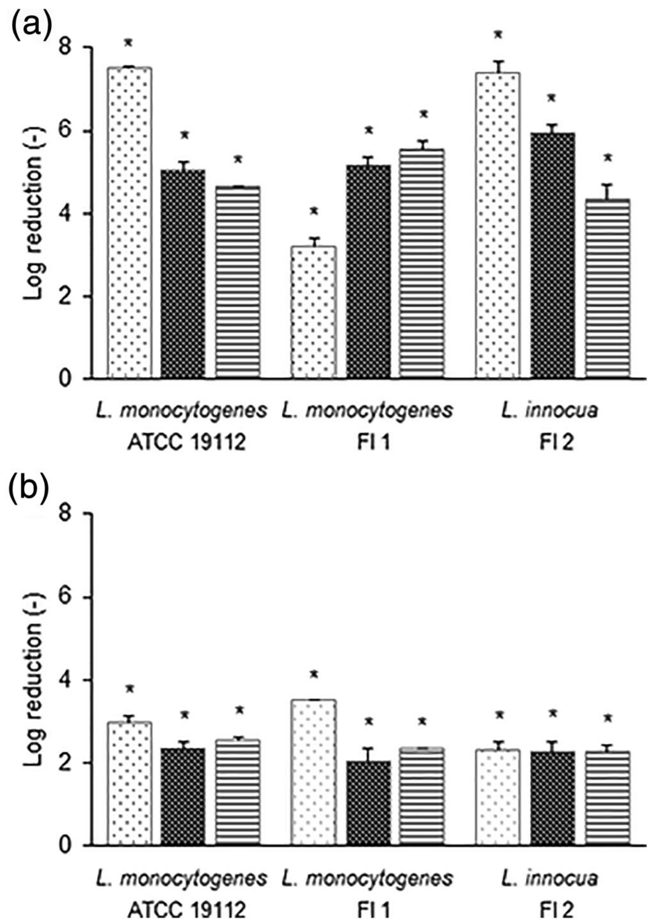

FIGURE 2 The effect of three disinfectants (QAC-, tertiary alkyl amine-, and chlorine-based) on inactivation of *Listeria* spp. in 24-hr (a) and 72-hr (b) biofilms after treatment for 5 min. The bars represent the mean values ± *SD* (n = 9). Asterisks indicate significant reduction as compared to the control (p < .05). QAC, quaternary ammonium compound

reduction in 72-hr biofilm; however, the difference in cell reduction between the 24-hr and 72-hr biofilm reached 0.3 log units and was statistically insignificant (p > .05).

The effectiveness of the three tested disinfectants revealed the QAC-based disinfectant to be the most effective against 24-hr biofilms of *L. monocytogenes* ATCC 19112 and *L. innocua* FI 2 (cell reductions by 7.5 and 7.4 log units, respectively), and the least effective against *L. monocytogenes* FI 1 biofilm (cell reduction by 3.2 log units). In the case of *L. monocytogenes* FI 1 biofilms, the chlorine-based disinfectant was the most effective (cell reduction by 5.6 log units), followed by *L. monocytogenes* ATCC and *L. innocua* FI 2 (cell reductions by 4.7 and 4.4 log units, respectively) (Figure 2a).

Cell reductions in 72-hr biofilms enabled observing that the QAC-based disinfectant was the most effective cell inactivator, as it caused cell reductions by 3.0, 3.5, and 2.3 log units in the case of *L. monocytogenes* ATCC 19112, *L. monocytogenes* FI 1, and *L. innocua* FI 2, respectively, and that the least effective was the tertiary alkyl amine-based one (cell reductions by 2.4, 2.0, and 2.3 log units, respectively) (Figure 2b). However, cell reductions did not exceed 3.0 log units for most disinfection treatments, except for the QAC-based disinfectant, which ensured *L. monocytogenes* FI 1 cell inactivation at the level of 3.5 log units.

Cell inactivation of *L. plantarum* strains in single species biofilms allowed observing their higher resistance as compared to *Listeria* spp. (Figure 3). Cell reductions in the 24-hr biofilm of *L. plantarum* PC 1.1 exposed to QAC-, tertiary alkyl amine-, and chlorine-based disinfectants reached 1.3, 1.2, and 1.0 log units, respectively. Cell reductions of *L. plantarum* PC 1.2 biofilms accounted for 0.6, 1.9, and 0.7 log units, whereas the inactivation of *L. plantarum* PC 3.2 biofilm resulted in cell reductions by 1.8, 0.7, and 2.2 log units, respectively. With only two exceptions, the inactivation of 72-hr biofilms resulted in higher values of cell reductions, which, however, did not exceed 2.2 log units and did not differ significantly from cell reductions achieved in the 24-hr biofilms ($p > .05$). A slightly higher resistance in 72-hr biofilms was observed for *L. plantarum* PC 1.2 exposed to the tertiary alkyl amine-based disinfectant (cell reduction by 1.9 and 1.4-log unit in 24-hr and 72-hr biofilms, respectively), and *L. plantarum* PC 3.2 exposed to the chlorine-based disinfectant (cell reduction by 2.2 and 2.1 log unit in 24-hr and 72-hr biofilms, respectively) (Figure 3). *L. plantarum* PC 1.2 was observed to be more resistant to the QAC- and chlorine-based disinfectants in both the 24-hr and 72-hr biofilms, compared to the other strains.

3.3 | Inactivation of individual strains in mixed species biofilms

Cell inactivation by the three disinfectants in the mixed species biofilms allowed observing that both *L. innocua* FI 2 and *L. plantarum* PC 1.2 were more resistant in the 72-hr than in the 24-hr biofilms with only one exception, that is, higher resistance did not occur for *L. innocua* FI 2 when the 72-hr biofilm was treated with the tertiary alkyl amine-based disinfectant (Figure 4). Again, cells in the 24-hr biofilms were the most susceptible to the QAC-based disinfectant, irrespective of strain. Cell reductions in the 24-hr biofilms reached 5.1, 0.6, and 0.4 for *L. innocua* FI 2 as well as 5.6, 3.0, and 0.6 for *L. plantarum* PC 1.2 treated with QAC-, tertiary alkyl amine-, and chlorine-based disinfectants, respectively. In turn, cell reductions observed in the 72-hr

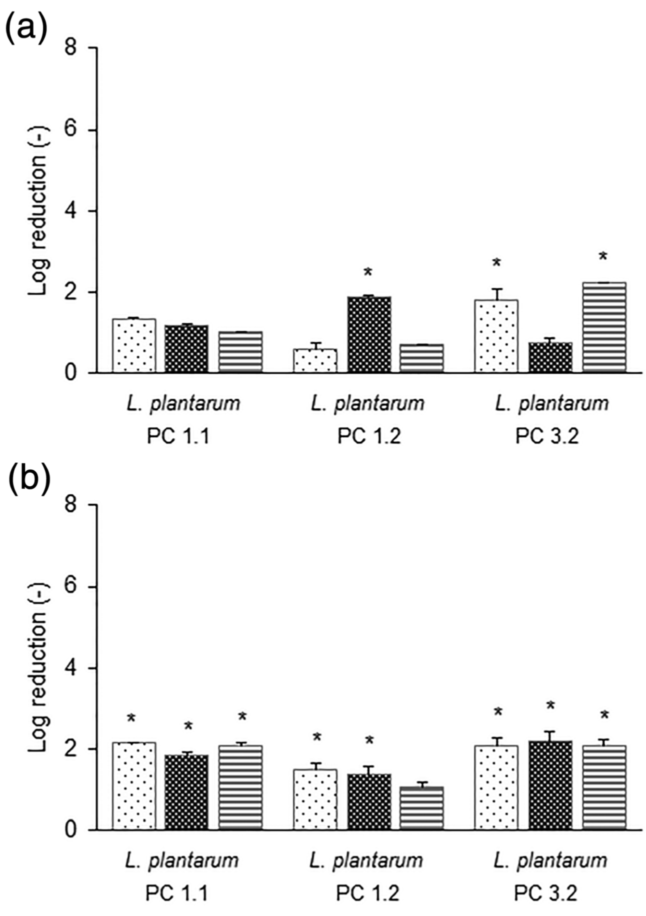

FIGURE 3 The effect of three disinfectants (QAC-, tertiary alkyl amine-, and chlorine-based) on inactivation of *L. plantarum* strains in 24-hr (a) and 72-hr (b) biofilms after treatment for 5 min. The bars represent the mean values ± standard deviations (n = 9). Asterisks indicate significant reduction as compared to the control ($p < .05$). QAC, quaternary ammonium compound

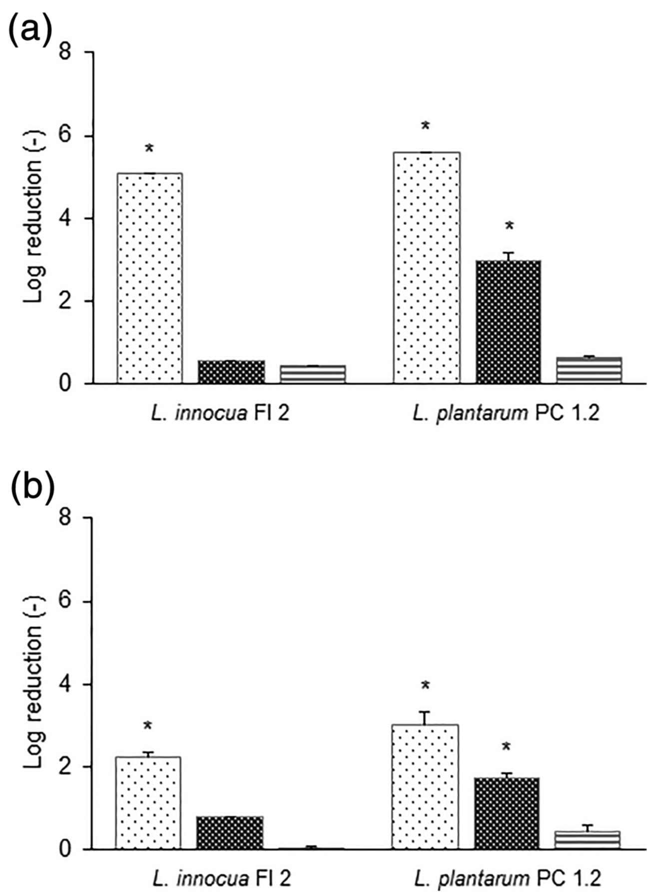

FIGURE 4 The effect of three disinfectants (QAC-, tertiary alkyl amine- and chlorine-based) on inactivation of *L. innocua* FI 2 and *L. plantarum* PC 1.2 in 24-hr (a) and 72-hr (b) mixed species biofilms after treatment for 5 min. The bars represent the mean values ± standard deviations ($n = 9$). Asterisks indicate significant reduction as compared to the control ($p < .05$). QAC, quaternary ammonium compound

biofilms treated with the tested disinfectants accounted for 2.2, 0.8, and 0.0 in the case of *L. innocua* FI 2 and for 3.0, 1.7, and 0.5 in the case of *L. plantarum* FI 2, respectively. In addition, *L. innocua* FI 2 cells were more resistant to all the three disinfectants in mixed species than in the single species biofilms, regardless of biofilm growth stage ($p < .05$), whereas *L. plantarum* PC 1.2 was more resistant only to the chlorine-based disinfectant; however, the difference was statistically significant only between 72-hr biofilms ($p < .05$).

3.4 | FISH and cell viability in mixed species biofilms

Fluorescent *in situ* hybridization of the 72-hr *L. innocua* FI 2–*L. plantarum* PC 1.2 mixed species biofilm in the LabTek system enabled observing blue (*L. innocua*) and red (*L. plantarum*) cells equally distributed in a biofilm as well as that the biofilm consisted of a dense structure of multiple layers of cells (Figure 5a). In turn, the LIVE/DEAD BacLight staining enabled differentiating live and dead cells of *L. innocua* and *L. plantarum* within the biofilms clearly showing a morphology of thinner and thicker cells of *L. innocua* and *L. plantarum*, respectively. Within the untreated biofilm, we found numerous live (green) and dead (red) cells and furthermore observed more damaged ones among *L. plantarum* (Figure 5b). Within the untreated biofilm too, the cells covered almost entire visual fields corresponding to eighth degree of surface colonization. Cell viability of biofilm cells when they were treated with the QAC-, tertiary alkyl-amine-, and chlorine-based disinfectants was

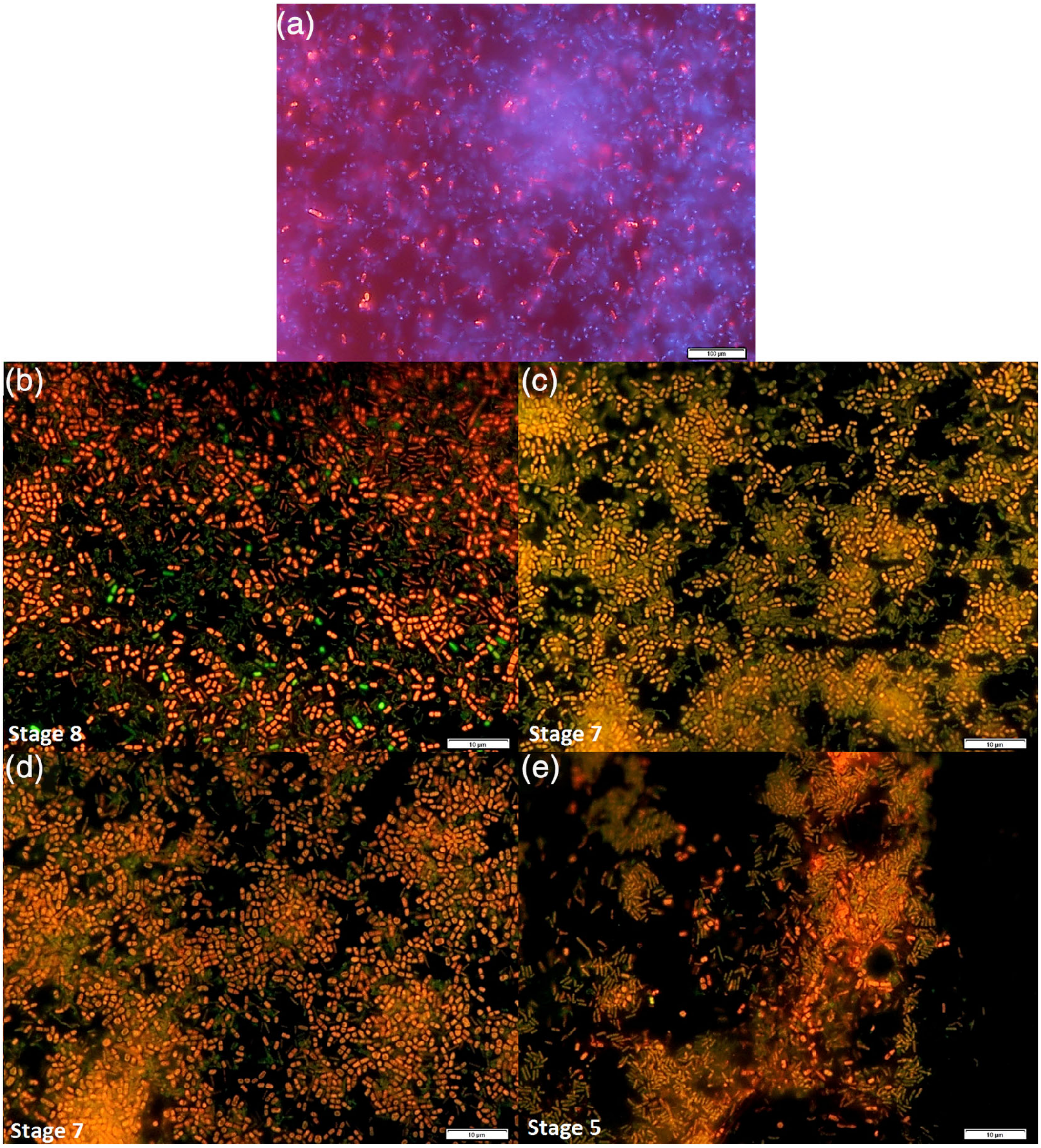

FIGURE 5 Representative EFM images of *L. innocua* FI 2 and *L. plantarum* PC 1.2 72-hr mixed species biofilms formed in LabTek system as described in Section 2 showing *Listeria* (blue) and *Lactobacillus* (red) cells visualized after FISH (a); and live (green) and dead (red) cells treated with different disinfectants and labeled with LIVE/DEAD BacLight (b—untreated, c—QAC-based, d—tertiary alkyl amine-based, and e—chlorine-based disinfection). QAC, quaternary ammonium compound

presented in Figure 5c–e. Importantly, we found that biofilms treated with both the QAC-based and the tertiary alkyl amine-based disinfectants maintained a dense structure of cell arrangements and that the major part of the visual fields were covered with the cells, which corresponded to the seventh degree of colonization (Figure 5b–d). In these biofilms, the number of dead cells increased, however not among *L. innocua* cells, suggesting their privileged localization in the biofilm and the shielding effect of *L. plantarum* cells toward *Listeria*. In addition, after the treatment with the tertiary alkyl amine-based disinfectant, the cells were distinctly red, whereas after exposure to the QAC-based disinfectant, they showed intermediate colors ranging from yellow to orange. In contrast, chlorine affected the mixed species biofilm differently (Figure 5e). Once exposed to chlorine, the biofilm structure was considerable less dense, the cells covered half the visual field, and the degree of colonization was determined at 5. Even though part of the cells retained their membrane integrity, biofilm structure was disrupted because cell aggregates were found to detach and almost no *L. plantarum* cells were observed.

4 | DISCUSSION

Bacterial biofilms contribute to problems with preserving hygiene in food processing plants where they pose the risk of food contamination by food-borne pathogens and food spoilage bacteria (Overney, 2017). Especially hazardous is *L. monocytogenes*, as it is able to persist within a food processing plant for a very long period of time; with the longest one established at 12 years (Wilks, Michels, & Keevil, 2006). One explanation for the long-term persistence of *Listeria* in the food processing environment is its ability to adhere and to form biofilms. In our study, the highest cell numbers of 8.6, 8.5, and 8.4 log CFU/ml after 24 hr under single-species conditions were scored for *L. monocytogenes* ATCC 19112, *L. monocytogenes* FI 1, and *L. innocua* FI 2, respectively, as well as *L. innocua* highly contributed to mixed-species biofilms reaching the level of 7.4 log CFU/ml of *L. innocua* FI 2. Simultaneously, the companion species, *L. plantarum* reached 7.3 log CFU/ml. The mixed species biofilms of *L. monocytogenes* and *L. plantarum* were also investigated by Van der Veen and Abee (2011). In their study, the biofilm was formed on the surface of 12-well microtiter plates in BHI, BHI-Mn, and BHI-Mn-glucose used as culture media. It was demonstrated that the use of the BHI supplemented with Mn resulted in the equal count of cells of both bacterial species in the biofilm (approximately 8.0 log CFU/well), which strongly corresponds with results of our study.

Biofilms are believed to be more resistant to different stress conditions, such as drying, UV radiation, and exposure to bactericides (Amalaradjou, Norris, & Venkitanarayanan, 2009; Borucki et al., 2003) including disinfectants used in the food industry (Aarnisalo, Lundén, Korkeala, & Wirtanen, 2007; Gram, Bagge-Ravn, Ng, Gymoese, & Vogel, 2007; Leriche, Chassaing, & Carpentier, 1999; Minei, Gomes, Regianne, D' Angelis, & De Martinis, 2008). As a result, cleaning and disinfection procedures being sufficient to inactive planktonic cells may appear less effective against biofilms. It is thus crucial to advance our understanding of formation of biofilms and their reaction to various chemical disinfectants and procedures commonly used in food processing plants, especially these formed by bacteria with pathogenic potential such as *Listeria*.

Chemical disinfection is often carried out with QAC-based disinfectants, which are representatives of the nonoxidizing chemicals (Langsrud & Sundheim, 1997; Gerba, 2015; Tezel & Pavlostathis, 2015). The principle of their action involves the impairment of the permeability of bacterial cell membrane, which results in the leakage of low-molecular-weight intracellular substances and in the coagulation of cytoplasmic proteins, which in turn contributes to the inhibition of the primary vital functions and eventually leads to cell death (McDonnell & Russell, 1999). One of the QAC-based disinfectants widely used in the food industry is the benzalkonium chloride (BAC), which contains a mixture of alkyl molecules with a chain length from C12 to C16 (Møretrø et al., 2017). Because of the wide spread usage of QACs, *L. monocytogenes* is believed to develop resistance to these antimicrobials by the expression of membrane active efflux pumps (Kovacevic et al., 2015; Martínez-Suárez, Ortiz, & López-Alonso, 2016; Romanova, Wolffs, Brovko, & Griffiths, 2006) and by the modification of fatty acid composition in cell membranes (To, Favrin, Romanova, & Griffiths, 2002). Another antimicrobial compound commonly used in the food processing plants is chlorine. Its widespread use is due to the easiness of its preparation and application and to its low price (Akbas, 2009; Schmidt, 2009; Van Houdt & Micjiels, 2010). The chlorine-based disinfectants release active chlorine and exhibit a high oxidative activity, owing to which they are able to damage proteins and lipoproteins of bacterial cells. Owing to their oxidative activity, they are capable of damaging the three-dimensional structure of a biofilm by eliminating EPSs and cause physical damage to the biofilm, which makes them instantly effective.

In this study, we showed a higher resistance of 72-hr than 24-hr single species biofilm of *Listeria* sp. to the tested disinfectants. In turn, the single species biofilms of *L. plantarum* showed that cells in 24-hr old biofilms were more resistant than in the 72-hr ones. Previous study reported that cells in older biofilms were more resistant than in earlier stages of biofilm formation (Liu, Yang, Du, & Pang, 2017; Yang, Kendall, Medeiros, & Sofos, 2009). It thus indicate the importance of the shape and three-dimensional organization of cells accompanied by EPS protective (Uhlich, Cooke, & Solomo, 2006) and shielding effect of dead cells (de Oliveira, Brugneraa, Cardosob, Alvesc, & Piccolia, 2010) in biofilm resistance. *L. plantarum* has a multilayered peptidoglycan cell wall with teichoic acids, EPSs, proteinaceous filaments (pili) and proteins, which may also protect against stress conditions such as disinfectants, regardless of stage of biofilm development (Sengupta et al., 2013).

Many previous studies have indicated *Listeria* to exhibit a poor ability for biofilm formation (Mosquera-Fernández, Rodríguez-López, Cabo, & Balsa-Canto, 2014); however, strains with strong capacity to adhere to the surface are known. de Oliveira et al. (2010) showed that

L. monocytogenes quickly adhered to the surface and then the cells proliferated slowly. Reis-Teixeira, Alves, and de Martinis (2017) showed that *L. monocytogenes* formed not very dense biofilm with small aggregates and microcolonies, in a honeycomb-like architecture on stainless steel and glass. It suggest that biofilms architecture can play an important role in biofilm resistance to antimicrobial agents. For mixed species biofilm of *L. innocua* FI 2 with *L. plantarum* PC 1.2, we showed that both strains were more resistant to all tested disinfectants than in single species, which concurs with previous work performed by, for example, Saá Ibusquiza, Cabo, and Herrera (2010), who showed the increased resistance of *L. monocytogenes* to BAC when grown in a co-culture with *Pseudomonas putida* on the stainless steel surface. In another study, *L. monocytogenes* exhibited an increased resistance to BAC when grown in the mixed species biofilms with *Escherichia coli* (Rodríguez-López, Carballo-Justo, Draper, & Cabo, 2017). Many studies suggest that interspecies interaction caused higher resistance of strains in mixed species biofilms than in single species (Millezi et al., 2012), but there are known also antagonistic interactions (Boari, Alves, Reis, Savian, & Piccoli, 2009; Rao, Webb, & Kjelleberg, 2005). It highlight that the development and resistance of biofilm depend of different aspects and more and more studied are needed to better understand the interactions between strains in mixed species biofilms. Furthermore, our results also showed that both strains in mixed species biofilms were more resistant in 72-hr biofilms. Thus, again demonstrating the importance of biofilm structure, maturity, and microbial interaction between strains in biofilms resistance.

In our study, we have also visualized the mixed-species biofilms by EFM with FISH and LIVE/DEAD staining. Results of FISH in the LabTek system showed that the mixed species biofilm of *L. innocua* with *L. plantarum* formed a dense and multilayer structure of cells almost evenly contributing to the community. Results obtained with using LIVE/DEAD staining showed live and dead cells of both strains and a morphology of thinner and thicker cells of *L. innocua* and *L. plantarum*, respectively. Image analysis suggests that *L. innocua* was protected by *L. plantarum*, as more dead cells were observed within lactobacilli population.

5 | CONCLUSION

In conclusion, *Listeria* spp. grown under single species conditions were more resistant to the tested disinfectants in 72-hr than in 24-hr biofilms, whereas *L. plantarum* strains showed no significant differences in disinfectant resistance between 24-hr and 72-hr biofilms. What is more, *L. innocua* FI 2 revealed increased resistance to all disinfection treatments in the mixed species biofilm, indicating a protective effect of the presence of *L. plantarum* PC 1.2. The biofilm formation and reaction to disinfectants, microscopically verified using FISH and LIVE/DEAD staining, showed that *L. innocua* and *L. plantarum* form a dense mixed biofilm and also suggested the shielding effect of *L. plantarum* on *L. innocua* in the mixed species biofilm. As a result, it needs to be highlighted that species interactions resulting in the enhanced resistance of microbes, which are introduced into the food processing environment may decrease the effectiveness of applied disinfection treatments and, therefore, a need emerges for searching other, more effective strategies for the elimination of biofilms formed by food pathogens and food spoiling microorganisms.

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CONFLICTS OF INTEREST

The authors declare no conflict of interest.

AUTHOR CONTRIBUTIONS

A.M.K. and M.A.O. conceived and planned the experiments. A.M.K. carried out the experiments, performed data collection and analysis, interpreted the data and wrote the manuscript. M.A.O. coordinated the study, analyzed and interpreted the data and wrote the manuscript. Both authors approved the final version of the manuscript.

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